# Topological Dirac Nodal Line Magnons in Layered Collinear Antiferromagnets and the Role of Interlayer Spin Canting Interaction

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**Abstract—**This work presents a comprehensive theoretical analysis of the symmetry-protected topological Dirac Nodal Line (DNL) magnonic phase in layered honeycomb collinear antiferromagnets, based on the foundational model by S. A. Owerre [EPL 122, 34001 (2019)][1]. We provide an explicit derivation of the magnon dispersion relations by bosonizing the spin Hamiltonian via the Holstein-Primakoff transformation and subsequent Bogoliubov diagonalization. The analysis emphasizes the crucial role of combined time-reversal and space-inversion (PT) symmetry in protecting the nodal lines, which exist where the bands touch. We rigorously define the quantized $\pi$ Berry phase as the key topological invariant for momentum-space loops enclosing the DNLs and explain why the consequent vanishing of the Berry curvature throughout the Brillouin zone leads to the absence of the magnon spin Nernst effect in this phase. Furthermore, this study extends the original model to include a next-nearest-neighbor (NNN) interlayer spin-canting interaction. By analytically deriving the new magnon energy spectrum, we demonstrate that this interaction breaks the protecting PT symmetry, lifting the band degeneracy and opening a gap, thereby destroying the topological DNL phase.

**Keywords—***Dirac nodal lines, Magnons, Honeycomb antiferromagnets, Berry Phase, PT Symmetry, Interlayer Spin Canting*

## 1. Lattice Structure and Reciprocal Space

To understand the magnon band structure, it is essential to first define the geometry of the crystal lattice in both real space and reciprocal space. The unique features of the energy bands are intrinsically linked to the symmetries of this underlying structure.

### 1.1. Real-Space Lattice

The system consists of two-dimensional (2D) honeycomb lattices stacked congruently along the z-axis.

#### *1.1.1. The 2D Honeycomb Lattice*

Each layer is a honeycomb lattice, which is not a Bravais lattice. It is a triangular Bravais lattice with a two-atom basis. We can define the primitive lattice vectors as:

$$\mathbf{a}_1 = (\sqrt{3}a, 0, 0) \tag{1}$$

$$\mathbf{a}_2 = \left(\frac{\sqrt{3}}{2}a, \frac{3}{2}a, 0\right) \tag{2}$$

where $a$ is the nearest-neighbor distance. The two atoms in the basis (sublattices A and B) are separated by a nearest-neighbor vector, for example, $\boldsymbol{\delta}_1 = (0, a, 0)$.

#### *1.1.2. The 3D Stacked Lattice*

The 3D crystal is formed by stacking these honeycomb layers directly on top of each other, separated by a distance $c$. The third primitive lattice vector is therefore:

$$\mathbf{a}_3 = (0, 0, c) \tag{3}$$

### 1.2. Finding the Reciprocal Space Basis

The reciprocal basis vectors for momentum space $(\vec{b}_1, \vec{b}_2, \vec{b}_3)$ are derived from the real-space primitive vectors $(\vec{a}_1, \vec{a}_2, \vec{a}_3)$ using the following standard definitions:

$$\vec{b}_1 = 2\pi \frac{\vec{a}_2 \times \vec{a}_3}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)} \quad , \quad \vec{b}_2 = 2\pi \frac{\vec{a}_3 \times \vec{a}_1}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

$$\vec{b}_3 = 2\pi \frac{\vec{a}_1 \times \vec{a}_2}{\vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)}$$

The denominator, $V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3)$, represents the volume of the real-space primitive unit cell.

#### *Calculation Steps*

We start with the given real-space vectors:

$$\vec{a}_1 = (\sqrt{3}a, 0, 0)$$

$$\vec{a}_2 = \left(\frac{\sqrt{3}}{2}a, \frac{3}{2}a, 0\right)$$

$$\vec{a}_3 = (0, 0, c)$$

**1. Calculate the Volume (Denominator)** First, we compute the cross product $\vec{a}_2 \times \vec{a}_3$:

$$\vec{a}_2 \times \vec{a}_3 = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \frac{\sqrt{3}}{2}a & \frac{3}{2}a & 0 \\ 0 & 0 & c \end{vmatrix} = \left(\frac{3}{2}ac\right)\hat{\mathbf{i}} - \left(\frac{\sqrt{3}}{2}ac\right)\hat{\mathbf{j}} = \left(\frac{3}{2}ac, -\frac{\sqrt{3}}{2}ac, 0\right)$$

Next, we take the dot product with $\vec{a}_1$ to find the volume $V$:

$$V = \vec{a}_1 \cdot (\vec{a}_2 \times \vec{a}_3) = (\sqrt{3}a, 0, 0) \cdot \left(\frac{3}{2}ac, -\frac{\sqrt{3}}{2}ac, 0\right) \tag{4}$$

$$= (\sqrt{3}a)\left(\frac{3}{2}ac\right) \tag{5}$$

$$= \frac{3\sqrt{3}}{2}a^2c \tag{6}$$

**2. Calculate the Reciprocal Vectors** Now we substitute the volume and the necessary cross products into the definitions.

For $\vec{b}_1$:

$$\vec{b}_1 = 2\pi \frac{(\frac{3}{2}ac, -\frac{\sqrt{3}}{2}ac, 0)}{\frac{3\sqrt{3}}{2}a^2c} = \frac{2\pi}{a}\frac{(\frac{3}{2}, -\frac{\sqrt{3}}{2}, 0)}{\frac{3\sqrt{3}}{2}} = \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)$$

For $\vec{b}_2$: We first need the cross product $\vec{a}_3 \times \vec{a}_1$:

$$\vec{a}_3 \times \vec{a}_1 = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ 0 & 0 & c \\ \sqrt{3}a & 0 & 0 \end{vmatrix} = (\sqrt{3}ac)\hat{\mathbf{j}} = (0, \sqrt{3}ac, 0)$$

$$\vec{b}_2 = 2\pi \frac{(0, \sqrt{3}ac, 0)}{\frac{3\sqrt{3}}{2}a^2c} = \frac{2\pi}{a}\frac{(0, \sqrt{3}, 0)}{\frac{3\sqrt{3}}{2}} = \frac{2\pi}{a}\left(0, \frac{2}{3}, 0\right)$$

For $\vec{b}_3$: We first need the cross product $\vec{a}_1 \times \vec{a}_2$:

$$\vec{a}_1 \times \vec{a}_2 = \begin{vmatrix} \hat{\mathbf{i}} & \hat{\mathbf{j}} & \hat{\mathbf{k}} \\ \sqrt{3}a & 0 & 0 \\ \frac{\sqrt{3}}{2}a & \frac{3}{2}a & 0 \end{vmatrix} = \left(\frac{3\sqrt{3}}{2}a^2\right)\hat{\mathbf{k}} = \left(0, 0, \frac{3\sqrt{3}}{2}a^2\right)$$

$$\vec{b}_3 = 2\pi \frac{(0, 0, \frac{3\sqrt{3}}{2}a^2)}{\frac{3\sqrt{3}}{2}a^2c} = \frac{2\pi}{c}(0, 0, 1)$$

#### Result

The final reciprocal basis vectors are:

$$\vec{b}_1 = \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)$$
$$\vec{b}_2 = \frac{2\pi}{a}\left(0, \frac{2}{3}, 0\right)$$
$$\vec{b}_3 = \frac{2\pi}{c}(0, 0, 1)$$

### 1.3. High-Symmetry Points in Reciprocal Space

High-symmetry points are specific locations of interest within the first Brillouin zone (the primitive unit cell of the reciprocal lattice). The paths connecting these points are used to plot the electronic band structure and phonon dispersion relations of the crystal, as the extrema of energy bands (maxima or minima) typically occur at these locations. For the simple hexagonal lattice, the Brillouin zone is a hexagonal prism.

The standard high-symmetry points are defined by fractional coordinates in the basis of the reciprocal lattice vectors $\vec{b}_1$, $\vec{b}_2$, and $\vec{b}_3$. We will now find their explicit Cartesian coordinates using the basis we previously derived:

$$\vec{b}_1 = \frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)$$
$$\vec{b}_2 = \frac{2\pi}{a}\left(0, \frac{2}{3}, 0\right)$$
$$\vec{b}_3 = \frac{2\pi}{c}(0, 0, 1)$$

#### 1.3.1. $\Gamma$ Point (Center)

The $\Gamma$ point is the origin of the first Brillouin zone.

$$\Gamma = 0\vec{b}_1 + 0\vec{b}_2 + 0\vec{b}_3$$
$$\Gamma = (0, 0, 0)$$

#### 1.3.2. $M$ Point (Edge Midpoint)

The $M$ point is the midpoint of a basal edge of the hexagonal prism.

$$M = \frac{1}{2}\vec{b}_1 + 0\vec{b}_2 + 0\vec{b}_3 = \frac{1}{2}\left(\frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)\right)$$
$$M = \frac{\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)$$

#### 1.3.3. $K$ and $K'$ Points (Basal Corners)

The $K$ and $K'$ points are the corners of the hexagonal face. They are inequivalent due to the two-atom basis of the honeycomb lattice.

$$K = \frac{1}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2 + 0\vec{b}_3 = \frac{1}{3}\left(\frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)\right) + \frac{1}{3}\left(\frac{2\pi}{a}\left(0, \frac{2}{3}, 0\right)\right)$$
$$K = \frac{2\pi}{3a}\left(\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right) + \left(0, \frac{2}{3}, 0\right)\right) = \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, \frac{1}{3}, 0\right)$$

Similarly for $K'$:

$$K' = \frac{2}{3}\vec{b}_1 - \frac{1}{3}\vec{b}_2 + 0\vec{b}_3 = \frac{2}{3}\left(\frac{2\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right)\right) - \frac{1}{3}\left(\frac{2\pi}{a}\left(0, \frac{2}{3}, 0\right)\right)$$
$$K' = \frac{2\pi}{3a}\left(\left(\frac{2}{\sqrt{3}}, -\frac{2}{3}, 0\right) - \left(0, \frac{2}{3}, 0\right)\right) = \frac{2\pi}{3a}\left(\frac{2}{\sqrt{3}}, -\frac{4}{3}, 0\right)$$

#### 1.3.4. $A$ Point (Top Face Center)

The $A$ point is the center of the top hexagonal face of the Brillouin zone.

$$A = 0\vec{b}_1 + 0\vec{b}_2 + \frac{1}{2}\vec{b}_3 = \frac{1}{2}\left(\frac{2\pi}{c}(0, 0, 1)\right)$$
$$A = \frac{\pi}{c}(0, 0, 1)$$

#### 1.3.5. $L$ Point (Vertical Edge Midpoint)

The $L$ point is the midpoint of a vertical edge connecting the top and bottom faces.

$$L = \frac{1}{2}\vec{b}_1 + 0\vec{b}_2 + \frac{1}{2}\vec{b}_3 = M + A$$
$$L = \frac{\pi}{a}\left(\frac{1}{\sqrt{3}}, -\frac{1}{3}, 0\right) + \frac{\pi}{c}(0, 0, 1) = \left(\frac{\pi}{a\sqrt{3}}, -\frac{\pi}{3a}, \frac{\pi}{c}\right)$$

#### 1.3.6. $H$ Point (Top Corner)

The $H$ point is a corner of the top hexagonal face of the Brillouin zone.

$$H = \frac{1}{3}\vec{b}_1 + \frac{1}{3}\vec{b}_2 + \frac{1}{2}\vec{b}_3 = K + A$$
$$H = \frac{2\pi}{3a}\left(\frac{1}{\sqrt{3}}, \frac{1}{3}, 0\right) + \frac{\pi}{c}(0, 0, 1) = \left(\frac{2\pi}{3a\sqrt{3}}, \frac{2\pi}{9a}, \frac{\pi}{c}\right)$$

### 1.4. Crystallographic (Miller) Indices

Notations like (010) and (001) are **Miller indices**, which are used in crystallography to define planes and surfaces within a crystal lattice.

#### 1.4.1. Definition

A plane is denoted by $(hkl)$, where $h, k, l$ are integers. They are found by a three-step process:

1. Find the intercepts of the plane with the crystal axes defined by the primitive vectors $\mathbf{a}_1, \mathbf{a}_2, \mathbf{a}_3$.
2. Take the reciprocal of these intercepts.
3. Multiply by a common factor to reduce them to the smallest possible integers.

#### 1.4.2. Interpretation in the Paper

- **(001) Surface BZ:** A plane with index (001) is parallel to the $\mathbf{a}_1$-$\mathbf{a}_2$ plane (the honeycomb plane) and intercepts the $\mathbf{a}_3$ axis (the z-axis). The "(001) surface BZ" mentioned in Fig. 3 is the 2D Brillouin zone of a single honeycomb layer.
- **Projected (010)-surface:** A plane with index (010) is parallel to the $\mathbf{a}_1$-$\mathbf{a}_3$ plane (the xz-plane) and intercepts the $\mathbf{a}_2$ axis. The "(010)-surface" refers to a termination of the 3D crystal along a plane perpendicular to the y-direction. The plot in Fig. 3(a) shows the allowed magnon energies for a finite slab of the material with such a surface. The "projection" means we are seeing all the allowed bulk states, plus any special states that are localized on that surface.

## 2. The Next-Nearest-Neighbor DMI Interaction

The Dzyaloshinskii-Moriya interaction (DMI) is an antisymmetric exchange interaction that arises from spin-orbit coupling in crystal structures lacking a center of inversion symmetry. Unlike the standard Heisenberg exchange which favors collinear spin alignments, the DMI energetically favors a canting (tilting) of neighboring spins.

### 2.1. Mathematical Formulation

The general form of the DMI Hamiltonian is:

$$\mathcal{H}_{DM} = \sum_{\langle\langle ij \rangle\rangle} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j) \tag{7}$$

where the sum is over next-nearest-neighbor (NNN) pairs of spins, $\mathbf{S}_i$ and $\mathbf{S}_j$. The vector $\mathbf{D}_{ij}$ is the DM vector, whose direction and

magnitude are determined by the crystal's symmetry, as dictated by Moriya's rules.

For the honeycomb lattice, there is no inversion center between two NNN sites (which are on the same sublattice). This broken symmetry allows for a DM vector that is perpendicular to the honeycomb plane. Therefore, we can write:

$$\mathbf{D}_{ij} = D\nu_{ij}\hat{z} \tag{8}$$

Here, $D$ is the magnitude of the DMI strength, and $\nu_{ij} = \pm 1$ is a sign factor that depends on the orientation of the bond connecting the NNN sites.

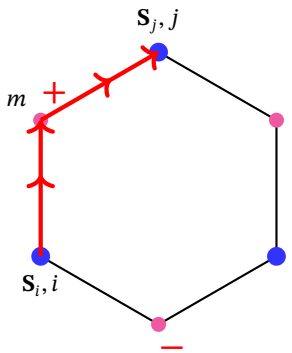


**Figure 1.** The NNN DMI path between two sites $i$ and $j$ on the same sublattice (blue). The sign $\nu_{ij}$ is determined by the intermediate site $m$ (magenta). For a counter-clockwise path $i \to m \to j$, $\nu_{ij} = +1$, while for a clockwise path, it would be $-1$.

### 2.2. The Sign Convention $\nu_{ij}$

The sign factor $\nu_{ij}$ is crucial and arises from the local environment. As illustrated in Fig. 1, the interaction between two NNN spins, $\mathbf{S}_i$ and $\mathbf{S}_j$, is mediated by their common nearest-neighbor on the other sublattice, site $m$. The sign of $\nu_{ij}$ can be determined by the "handedness" of the path from $i$ to $j$ via $m$. A common convention is:

- If the path $i \to m \to j$ is counter-clockwise, $\nu_{ij} = +1$.
- If the path is clockwise, $\nu_{ij} = -1$.

This results in an alternating pattern of signs for the DMI on the NNN bonds around any given hexagon, which is what gives rise to the complex momentum dependence of the $f^D(\mathbf{k}_{||})$ term in the Hamiltonian. The term $\mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j)$ is minimized when the spins cant in a way that makes their cross product anti-parallel to the DM vector. This interaction is the microscopic origin of many interesting topological phenomena in magnonics.

## 3. Introduction: The Spin Hamiltonian

We begin with the microscopic spin Hamiltonian for a layered honeycomb antiferromagnet in the absence of an external magnetic field ($\mathbf{H} = 0$). The model includes intralayer and interlayer exchange interactions, and the Dzyaloshinskii-Moriya (DM) interaction, and is described by the Hamiltonian[1]:

$$\mathcal{H} = J \sum_{\langle ij \rangle, l} \mathbf{S}_{i,l} \cdot \mathbf{S}_{j,l} + \sum_{\langle\langle ij \rangle\rangle, l} \mathbf{D}_{ij,l} \cdot (\mathbf{S}_{i,l} \times \mathbf{S}_{j,l}) + J_c \sum_{\langle ll' \rangle, i} \mathbf{S}_{i,l} \cdot \mathbf{S}_{i,l'}. \tag{9}$$

Here, $J > 0$ is the nearest-neighbor (NN) intralayer antiferromagnetic coupling, $\mathbf{D}_{ij,l}$ is the next-nearest-neighbor (NNN) DM interaction, and $J_c > 0$ is the NN interlayer antiferromagnetic coupling. The study of the low-energy excitations (magnons) of this system reveals a rich topological structure characterized by lines of band touching.

### 3.1. The Origin of PT Symmetry in the Collinear Antiferromagnet

The system possesses PT symmetry because the combined operation of Time-Reversal ($\mathcal{T}$) and space-Inversion ($\mathcal{P}$) leaves the Hamiltonian and its ground state unchanged, even though neither operation is a symmetry on its own. Let's analyze the effect of these operations on the collinear Néel ground state, where spins on sublattice A point up ($\mathbf{S}_A = S\hat{z}$) and spins on sublattice B point down ($\mathbf{S}_B = -S\hat{z}$).

#### 3.1.1. Action of Individual Symmetries ($\mathcal{T}$ and $\mathcal{P}$)

First, we consider the effect of each operation individually.

- **Time-Reversal ($\mathcal{T}$):** This operation reverses all magnetic moments (spins).
$$\mathcal{T}: \quad \mathbf{S} \to -\mathbf{S} \tag{10}$$
Applying this to our ground state:
$$\mathcal{T}(\mathbf{S}_A) = -S\hat{z}$$
$$\mathcal{T}(\mathbf{S}_B) = +S\hat{z}$$
The resulting state has spins pointing down on sublattice A and up on sublattice B. This is a different magnetic configuration from the original state. Therefore, $\mathcal{T}$ **is not a symmetry** of the system.
- **Space-Inversion ($\mathcal{P}$):** For the honeycomb lattice, the inversion center lies on a bond, so the operation effectively swaps the two sublattices.
$$\mathcal{P}: \quad \text{sublattice A} \leftrightarrow \text{sublattice B} \tag{11}$$
Applying this to our ground state:
$$\mathcal{P}(\text{State}) \implies \text{Spins on A are now } \mathbf{S}_B = -S\hat{z}$$
$$\mathcal{P}(\text{State}) \implies \text{Spins on B are now } \mathbf{S}_A = +S\hat{z}$$
Again, the resulting state has spins pointing down on A and up on B. This is different from the original state. Therefore, $\mathcal{P}$ **is not a symmetry** of the system.

#### 3.1.2. Action of the Combined Symmetry ($\mathcal{PT}$)

Now, we apply both operations in succession to the initial ground state.

1. **Initial State:** Spins on A are $+S\hat{z}$; spins on B are $-S\hat{z}$.
2. **Apply $\mathcal{T}$ first:** The spins are flipped. The state becomes: spins on A are $-S\hat{z}$; spins on B are $+S\hat{z}$.
3. **Apply $\mathcal{P}$ to the new state:** The sublattices are swapped.
   - The spins that were on sublattice A (which are now $-S\hat{z}$) are moved to sublattice B.
   - The spins that were on sublattice B (which are now $+S\hat{z}$) are moved to sublattice A.
4. **Final State:** Spins on A are $+S\hat{z}$; spins on B are $-S\hat{z}$.

The final state is identical to the initial state. Since the combined operation ($\mathcal{PT}$) leaves the ground state invariant, we conclude that **PT is a symmetry** of the collinear antiferromagnet. This symmetry is what protects the Dirac Nodal Lines from being gapped by perturbations like the DMI, which also respects this combined symmetry.

### 3.2. Symmetry Analysis: DMI vs. External Magnetic Field

The topological protection of the Dirac Nodal Lines (DNLs) relies on the presence of PT symmetry. To understand the transition to the gapped Chern insulator phase, it is crucial to analyze how different terms in the Hamiltonian transform under the **Time-Reversal ($\mathcal{T}$)** and **space-Inversion ($\mathcal{P}$)** operations.

#### 3.2.1. Why the DMI Term Preserves PT Symmetry

The Dzyaloshinskii-Moriya interaction is given by:

$$\mathcal{H}_{DM} = \sum_{\langle\langle ij\rangle\rangle} \mathbf{D}_{ij} \cdot (\mathbf{S}_i \times \mathbf{S}_j) \tag{12}$$

To analyze its symmetry properties, we must consider how each component transforms.

- The spin operator $\mathbf{S}$ is an **axial vector** (or pseudovector).
- The DM vector $\mathbf{D}_{ij}$ is a **true vector** (or polar vector).
- The cross product of two axial vectors, $(\mathbf{S}_i \times \mathbf{S}_j)$, is also an axial vector.

**Action of space-Inversion ($\mathcal{P}$):** The inversion operation sends $\mathbf{r} \to -\mathbf{r}$. Under this transformation:

- True vectors flip sign: $\mathcal{P}(\mathbf{D}_{ij}) = -\mathbf{D}_{ij}$.
- Axial vectors do not change sign: $\mathcal{P}(\mathbf{S}_i \times \mathbf{S}_j) = (\mathbf{S}_i \times \mathbf{S}_j)$.

Applying this to the DM Hamiltonian term:

$$\mathcal{P}(\mathcal{H}_{DM}) = \sum \mathcal{P}(\mathbf{D}_{ij})\cdot\mathcal{P}(\mathbf{S}_i\times\mathbf{S}_j) = \sum(-\mathbf{D}_{ij})\cdot(\mathbf{S}_i\times\mathbf{S}_j) = -\mathcal{H}_{DM} \tag{13}$$

Thus, the DM interaction **breaks space-inversion symmetry ($\mathcal{P}$)**.

**Action of Time-Reversal ($\mathcal{T}$):** The time-reversal operation sends $t \to -t$. Under this transformation:

- Spins flip sign: $\mathcal{T}(\mathbf{S}_i) = -\mathbf{S}_i$.
- The DM vector, arising from spin-orbit coupling, is odd under time-reversal: $\mathcal{T}(\mathbf{D}_{ij}) = -\mathbf{D}_{ij}$.
- The cross product of spins transforms as: $\mathcal{T}(\mathbf{S}_i \times \mathbf{S}_j) = (-\mathbf{S}_i) \times (-\mathbf{S}_j) = (\mathbf{S}_i \times \mathbf{S}_j)$.

Applying this to the DM Hamiltonian term:

$$\mathcal{T}(\mathcal{H}_{DM}) = \sum \mathcal{T}(\mathbf{D}_{ij})\cdot\mathcal{T}(\mathbf{S}_i\times\mathbf{S}_j) = \sum(-\mathbf{D}_{ij})\cdot(\mathbf{S}_i\times\mathbf{S}_j) = -\mathcal{H}_{DM} \tag{14}$$

Thus, the DM interaction also **breaks time-reversal symmetry ($\mathcal{T}$)**.

**Action of Combined PT Symmetry:** Now we apply both operations in succession:

$$\mathcal{P}\mathcal{T}(\mathcal{H}_{DM}) = \mathcal{P}(\mathcal{T}(\mathcal{H}_{DM})) = \mathcal{P}(-\mathcal{H}_{DM}) = -(\mathcal{P}(\mathcal{H}_{DM})) = -(-\mathcal{H}_{DM}) = +\mathcal{H}_{DM} \tag{15}$$

The DM interaction is **invariant under the combined PT operation**. Because it respects the protecting symmetry of the DNL phase, it cannot open a gap in the magnon spectrum.

#### 3.2.2. Why the External Magnetic Field Breaks PT Symmetry

The Zeeman term from the external magnetic field is given by:

$$\mathcal{H}_H = -H \sum_i S_i^z \tag{16}$$

**Action of space-Inversion ($\mathcal{P}$):** Under inversion, the z-component of an axial vector is invariant: $\mathcal{P}(S_i^z) = S_i^z$. The sum over all sites is also unchanged.

$$\mathcal{P}(\mathcal{H}_H) = -H \sum_i \mathcal{P}(S_i^z) = -H \sum_i S_i^z = \mathcal{H}_H \tag{17}$$

Thus, the Zeeman term **preserves space-inversion symmetry ($\mathcal{P}$)**.

**Action of Time-Reversal ($\mathcal{T}$):** Under time-reversal, spins flip: $\mathcal{T}(S_i^z) = -S_i^z$.

$$\mathcal{T}(\mathcal{H}_H) = -H \sum_i \mathcal{T}(S_i^z) = -H \sum_i(-S_i^z) = +H \sum_i S_i^z = -\mathcal{H}_H \tag{18}$$

Thus, the Zeeman term **breaks time-reversal symmetry ($\mathcal{T}$)**.

**Action of Combined PT Symmetry:** Now we apply both operations in succession:

$$\mathcal{P}\mathcal{T}(\mathcal{H}_H) = \mathcal{P}(\mathcal{T}(\mathcal{H}_H)) = \mathcal{P}(-\mathcal{H}_H) = -(\mathcal{P}(\mathcal{H}_H)) = -(\mathcal{H}_H) = -\mathcal{H}_H \tag{19}$$

The Zeeman term is **not invariant under the combined PT operation**. Because it explicitly breaks the symmetry that protects the DNLs, the external magnetic field acts as a perturbation that lifts the band degeneracy and opens the topological gap, driving the system into the Magnon Chern Insulator phase.

## 4. Hamiltonian Bosonization and Diagonalization

To analyze the magnon spectrum at low temperatures, we employ the Holstein-Primakoff (HP) transformation to express the spin operators in terms of bosonic creation and annihilation operators. This is followed by a Fourier transform and a Bogoliubov transformation to diagonalize the Hamiltonian.

### 4.1. Derivation of the Momentum-Space Hamiltonian

The derivation involves applying the HP and Fourier transformations to each term of Eq. (9) and retaining terms up to quadratic order in the bosonic operators.

#### 4.1.1. Holstein-Primakoff and Fourier Transformations

For the collinear Néel ordered state, we define separate transformations for the two sublattices A and B. For a site $i$ on sublattice A and a site $j$ on sublattice B:

$$S_{i,A}^z = S - a_i^\dagger a_i, \qquad S_{i,A}^+ \approx \sqrt{2S}a_i, \qquad S_{i,A}^- \approx \sqrt{2S}a_i^\dagger \tag{20}$$

$$S_{j,B}^z = -S + b_j^\dagger b_j, \qquad S_{j,B}^+ \approx \sqrt{2S}b_j^\dagger, \qquad S_{j,B}^- \approx \sqrt{2S}b_j \tag{21}$$

We then Fourier transform from real space operators to momentum space operators:

$$a_i = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}_i} a_{\mathbf{k}}, \quad b_j = \frac{1}{\sqrt{N}} \sum_{\mathbf{k}} e^{i\mathbf{k}\cdot\mathbf{r}_j} b_{\mathbf{k}}. \tag{22}$$

#### 4.1.2. Term-by-Term Bosonization

**Intralayer Exchange ($\mathcal{H}_J$):** The dynamic part arises from the $S^+S^-$ terms. For nearest-neighbor sites $i \in A$ and $j \in B$:

$$\frac{1}{2}(S_{i,A}^+ S_{j,B}^- + S_{i,A}^- S_{j,B}^+) \approx S(a_i b_j + a_i^\dagger b_j^\dagger). \tag{23}$$

Summing over nearest neighbors and Fourier transforming introduces a geometric factor $\gamma_{\mathbf{k}} = \sum_\delta e^{i\mathbf{k}\cdot\delta} = (1 + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_1} + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_2})$. This results in the term $JS\sum_{\mathbf{k}}(\gamma_{\mathbf{k}}^* a_{\mathbf{k}}^\dagger b_{-\mathbf{k}}^\dagger + \gamma_{\mathbf{k}} a_{\mathbf{k}} b_{-\mathbf{k}})$, which forms the off-diagonal component $f(\mathbf{k}_{||}) = JS\gamma_{\mathbf{k}}^*$ of the $\mathcal{B}(\mathbf{k})$ matrix.

**Interlayer Exchange ($\mathcal{H}_{J_c}$):** This term couples spins on the same sublattice in adjacent layers. The dynamic part for a site $i \in A$ is $\approx S(a_{i,l}a_{i,l'}^\dagger + a_{i,l}^\dagger a_{i,l'})$. After Fourier transforming and summing over adjacent layers ($l' = l \pm 1$), the pair creation/annihilation part forms $f(k_z) = 2J_c S\cos(k_z)$.

**DM Interaction ($\mathcal{H}_{DM}$):** This term is between next-nearest neighbors on the same sublattice. For sublattice A, $\mathcal{H}_{DM,A} \approx \frac{DS}{i}\sum_{\langle\langle ij\rangle\rangle}(a_i a_j^\dagger - a_i^\dagger a_j)$. Fourier transforming yields a diagonal term $\sum_{\mathbf{k}} \Delta_{\mathbf{k}} a_{\mathbf{k}}^\dagger a_{\mathbf{k}}$, where the geometric factor $\Delta_{\mathbf{k}}$ leads to the function $f^D(\mathbf{k}_{||})$.

#### 4.1.3. Assembling the Final Matrix

Combining all terms, the constant parts from $S^zS^z$ interactions and the diagonal parts of the exchange terms form the overall energy offset, $f_0 = 3JS + 2J_cS$. The final quadratic Hamiltonian is $\mathcal{H} = \frac{1}{2}\sum_{\mathbf{k}} \Psi^\dagger(\mathbf{k})\mathcal{H}(\mathbf{k})\Psi(\mathbf{k})$, with the basis $\Psi^\dagger(\mathbf{k}) =$

$(a^\dagger_{\mathbf{k},A}, a^\dagger_{\mathbf{k},B}, a_{-\mathbf{k},A}, a_{-\mathbf{k},B})$ and the matrix:

$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} \mathcal{A}(\mathbf{k}) & \mathcal{B}(\mathbf{k}) \\ \mathcal{B}^*(-\mathbf{k}) & \mathcal{A}^*(-\mathbf{k}) \end{pmatrix}, \tag{24}$$

where $\mathcal{A}(\mathbf{k}) = [f_0 + f^D(\mathbf{k}_{||})]I_{2\times 2}$ and $\mathcal{B}(\mathbf{k}) = \begin{pmatrix} f(k_z) & f(\mathbf{k}_{||}) \\ f^*(\mathbf{k}_{||}) & f(k_z) \end{pmatrix}$.

### 4.2. Detailed Bogoliubov Diagonalization

The Hamiltonian is diagonalized via a Bogoliubov transformation, which is equivalent to solving the generalized eigenvalue problem $\mathcal{H}(\mathbf{k})\Phi = E\sigma_z\Phi$, where $\sigma_z = \mathrm{diag}(1, 1, -1, -1)$. This can be rewritten as:

$$\begin{pmatrix} \mathcal{A}(\mathbf{k}) - EI & \mathcal{B}(\mathbf{k}) \\ \mathcal{B}^*(-\mathbf{k}) & \mathcal{A}^*(-\mathbf{k}) + EI \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix} = 0. \tag{25}$$

For a non-trivial solution, the determinant of the matrix must be zero. Using the formula for the determinant of a block matrix and noting that $\mathcal{A}$ is real, we get:

$$\det\left((\mathcal{A}(\mathbf{k})^2 - E^2 I) - \mathcal{B}(\mathbf{k})\mathcal{B}^\dagger(\mathbf{k})\right) = 0. \tag{26}$$

This implies that $E^2$ is related to the eigenvalues of $\mathcal{B}(\mathbf{k})\mathcal{B}^\dagger(\mathbf{k})$. The eigenvalues $\lambda_{\mathcal{B}}$ of $\mathcal{B}(\mathbf{k})$ are found by solving $\det(\mathcal{B}(\mathbf{k}) - \lambda I) = 0$:

$$(f(k_z) - \lambda)^2 - |f(\mathbf{k}_{||})|^2 = 0 \implies \lambda_{\mathcal{B},\pm} = f(k_z) \pm |f(\mathbf{k}_{||})|. \tag{27}$$

Since $\mathcal{B}(\mathbf{k})$ is a normal matrix, the eigenvalues of $\mathcal{B}(\mathbf{k})\mathcal{B}^\dagger(\mathbf{k})$ are simply $|\lambda_{\mathcal{B},\pm}|^2 = (f(k_z) \pm |f(\mathbf{k}_{||})|)^2$. Substituting this back into the characteristic equation gives:

$$(f_0 + f^D(\mathbf{k}_{||}))^2 - E^2 = (f(k_z) \pm |f(\mathbf{k}_{||})|)^2. \tag{28}$$

Solving for $E$ yields the final dispersion for the two magnon bands:

$$E_\pm(\mathbf{k}) = \sqrt{(f_0 + f^D(\mathbf{k}_{||}))^2 - (f(k_z) \pm |f(\mathbf{k}_{||})|)^2}. \tag{29}$$

## 5. The Dirac Nodal Line Phase

At zero magnetic field, the system exhibits a topological semi-metal phase. The bands become degenerate, $E_+(\mathbf{k}) = E_-(\mathbf{k})$, when the condition $f(k_z) \cdot |f(\mathbf{k}_{||})| = 0$ is met. This degeneracy does not occur at isolated points but along continuous lines in the Brillouin zone, known as **Dirac Nodal Lines (DNLs)**.

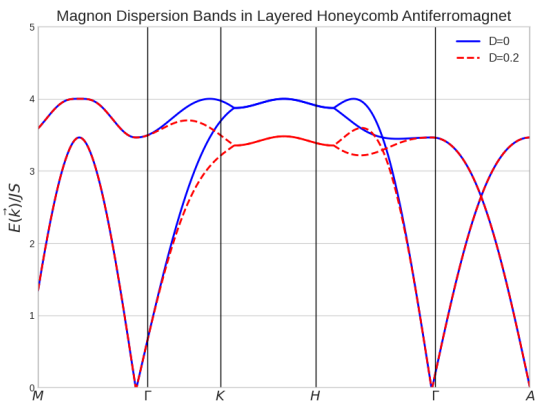


**Figure 2.** Magnon dispersion bands along high-symmetry lines in the Brillouin zone for $J_c/J = 0.5$. The solid blue lines correspond to $D = 0$, showing Dirac Nodal Lines (DNLs) where the bands touch. The dashed red lines ($D = 0.2$) show that the DNLs persist even with DMI.

## 6. Topology of the DNL Phase

The DNLs are not accidental degeneracies; they are topologically protected by the symmetries of the system. The topology of this gapless phase is characterized by a Berry phase, while other invariants like the Chern number are ill-defined.

### 6.1. PT Symmetry and its Role in DNL Topology

**PT symmetry** is a composite symmetry formed by the successive application of two fundamental operations: **Time-Reversal ($\mathcal{T}$)** and **space-Inversion ($\mathcal{P}$)**.

1. **Time-Reversal ($\mathcal{T}$):** This operation reverses the direction of time. In magnetism, this means it flips the direction of all spins and the momentum of all particles:

$$\mathcal{T}: \quad \mathbf{S} \to -\mathbf{S}, \quad \mathbf{k} \to -\mathbf{k} \tag{30}$$

2. **Space-Inversion ($\mathcal{P}$):** This operation inverts the system through a central point, effectively flipping the sign of all spatial coordinates:

$$\mathcal{P}: \quad \mathbf{r} \to -\mathbf{r} \tag{31}$$

In the DNL phase (a collinear antiferromagnet with $H = 0$), the Hamiltonian is invariant under the combined **PT** operation. While individual $\mathcal{T}$ or $\mathcal{P}$ operations might change the system, applying both in succession leaves it unchanged.

#### *Why PT Symmetry is Crucial for DNL Topology*

The invariance of the Hamiltonian under PT symmetry is what **protects** the Dirac Nodal Lines. It guarantees their existence and stability against perturbations.

- **Protection Against Gapping:** The band touching along the DNLs is not accidental. For a gap to open and destroy a nodal line, the perturbation added to the system **must break PT symmetry**. Any perturbation that preserves PT symmetry (like the DMI in this model) is incapable of gapping the nodal lines. This is the essence of "symmetry protection."
- **Vanishing Berry Curvature:** PT symmetry imposes a powerful mathematical constraint on the Berry curvature, $\mathbf{\Omega}(\mathbf{k})$. Symmetries dictate how the curvature transforms:
  - Time-reversal requires: $\mathbf{\Omega}(\mathbf{k}) = -\mathbf{\Omega}(-\mathbf{k})$
  - Inversion requires: $\mathbf{\Omega}(\mathbf{k}) = \mathbf{\Omega}(-\mathbf{k})$

  For a system that has both (i.e., is PT symmetric), the only way to satisfy both conditions simultaneously is for the Berry curvature to be **identically zero everywhere** in the Brillouin zone:

$$\mathbf{\Omega}(\mathbf{k}) = 0 \tag{32}$$

  This vanishing Berry curvature is a direct and measurable consequence of the protecting symmetry. It explains why the **magnon spin Nernst effect** and the **magnon thermal Hall effect**, which depend on a non-zero Berry curvature, are absent in the DNL phase.

### 6.2. Topological Invariant: The Berry Phase

For a gapless system like a DNL semi-metal, the primary topological invariant is the **Berry phase** (or Zak phase), $\gamma_C$, calculated along a closed loop $C$ in momentum space. It is defined as the line integral of the Berry connection, $\mathbf{A}_n(\mathbf{k})$:

$$\gamma_C = \oint_C \mathbf{A}_n(\mathbf{k}) \cdot d\mathbf{k}, \tag{33}$$

where the Berry connection for a band $n$ is given by:

$$\mathbf{A}_n(\mathbf{k}) = i\langle u_n(\mathbf{k})|\nabla_{\mathbf{k}}|u_n(\mathbf{k})\rangle, \tag{34}$$

with $|u_n(\mathbf{k})\rangle$ being the periodic part of the Bloch eigenvector for that band.

The key result is that if the loop $C$ **does not** enclose a nodal line, the Berry phase is trivial, $\gamma_C = 0$. However, if the loop $C$ **does** enclose a segment of a nodal line, the Berry phase is quantized to a

non-trivial value of $\gamma_C = \pm\pi$. This quantized, non-zero value acts as a topological charge, confirming that the nodal line is a topologically protected defect in momentum space.

### 6.3. Berry Curvature and the Chern Number in the DNL Phase

The Berry curvature, $\mathbf{\Omega}_n(\mathbf{k})$, is defined as the curl of the Berry connection:

$$\mathbf{\Omega}_n(\mathbf{k}) = \nabla_{\mathbf{k}} \times \mathbf{A}_n(\mathbf{k}). \tag{35}$$

In the DNL phase, the presence of PT symmetry imposes a strict constraint: the Berry curvature must be identically zero everywhere in the Brillouin zone, $\mathbf{\Omega}_n(\mathbf{k}) = 0$. This is because PT symmetry requires $\mathbf{\Omega}_n(\mathbf{k}) = -\mathbf{\Omega}_n(-\mathbf{k})$, while inversion symmetry (part of PT) requires $\mathbf{\Omega}_n(\mathbf{k}) = \mathbf{\Omega}_n(-\mathbf{k})$. The only way to satisfy both conditions is for the curvature to be zero.

The first **Chern number**, $\mathcal{C}_n$, is a topological invariant for gapped 2D systems, calculated by integrating a component of the Berry curvature over the 2D Brillouin zone:

$$\mathcal{C}_n = \frac{1}{2\pi} \int_{BZ} d^2k_{||}\, \Omega^n_{xy}(\mathbf{k}). \tag{36}$$

Since the Berry curvature is zero everywhere in the DNL phase, the Chern number is trivially zero. More fundamentally, the Chern number is only well-defined and quantized for systems with a finite energy gap throughout the Brillouin zone. As the DNL phase is gapless, the Chern number is not the appropriate invariant to describe its topology. The non-trivial topology is instead captured by the $\pi$ Berry phase associated with the nodal lines themselves.

### 6.4. Calculation of the Quantized $\pi$ Berry Phase

To demonstrate the topological nature of the DNL, we can calculate the Berry phase for a closed loop in momentum space that encircles the nodal line. The result is a quantized value of $\pm\pi$, which acts as a topological invariant. The calculation is most tractable by considering a low-energy effective Hamiltonian that captures the physics near the band-touching point.

#### 6.4.1. The Effective 2x2 Dirac Hamiltonian

Near a band-touching point, such as the K point, the complex 4x4 Bogoliubov-de Gennes Hamiltonian can be simplified. The low-energy dynamics of the two relevant bands are accurately described by an effective 2x2 Hamiltonian, which takes the form of a massive 2D Dirac Hamiltonian:

$$\mathcal{H}_{\text{eff}}(\mathbf{q}) = v(q_x\sigma_x + q_y\sigma_y) + \Delta\sigma_z = \begin{pmatrix} \Delta & v(q_x - iq_y) \\ v(q_x + iq_y) & -\Delta \end{pmatrix} \tag{37}$$

Here, $\mathbf{q} = \mathbf{k} - \mathbf{K}$ is the small momentum deviation from the K point, $v$ is the magnon velocity, $\sigma_{x,y,z}$ are the Pauli matrices, and $\Delta$ is a mass term that would open a gap. The Dirac Nodal Line itself corresponds to the case where $\mathbf{q} = 0$ and the gap is closed, i.e., $\Delta = 0$.

#### 6.4.2. Eigenstates and Berry Connection

The energy eigenvalues of this effective Hamiltonian are $E_\pm(\mathbf{q}) = \pm\sqrt{v^2q^2 + \Delta^2}$. The normalized eigenvector for the lower energy band, $|u_-(\mathbf{q})\rangle$, is found to be:

$$|u_-(\mathbf{q})\rangle = \frac{1}{\sqrt{2E(E-\Delta)}} \begin{pmatrix} -v(q_x - iq_y) \\ E - \Delta \end{pmatrix} \tag{38}$$

where $E = \sqrt{v^2q^2 + \Delta^2}$. The Berry connection, $\mathbf{A}(\mathbf{q}) = i\langle u_-|\nabla_{\mathbf{q}}|u_-\rangle$, can be calculated from this eigenvector. Its components are:

$$A_x(\mathbf{q}) = \frac{v^2}{2(v^2q^2 + \Delta^2)} \left(\frac{\Delta}{E}\right) q_y \tag{39}$$

$$A_y(\mathbf{q}) = -\frac{v^2}{2(v^2q^2 + \Delta^2)} \left(\frac{\Delta}{E}\right) q_x \tag{40}$$

#### 6.4.3. Berry Phase Calculation via Stokes' Theorem

The Berry phase, $\gamma_C$, is the line integral of the Berry connection around a closed loop $C$ enclosing the DNL (the origin, $\mathbf{q} = 0$). We use Stokes' theorem to convert this line integral into a surface integral of the Berry curvature, $\Omega_z$:

$$\gamma_C = \oint_C \mathbf{A}(\mathbf{q}) \cdot d\mathbf{q} = \iint_S (\nabla_{\mathbf{q}} \times \mathbf{A}) \cdot d\mathbf{S} = \iint_S \Omega_z\, dq_x dq_y \tag{41}$$

The Berry curvature is calculated as:

$$\Omega_z = \frac{\partial A_y}{\partial q_x} - \frac{\partial A_x}{\partial q_y} = -\frac{v^2\Delta}{2(v^2q^2 + \Delta^2)^{3/2}} \tag{42}$$

We integrate this curvature over a circular area of radius $R$ in momentum space. In polar coordinates ($q_x = q\cos\theta, q_y = q\sin\theta$), the integral becomes:

$$\gamma_C = \int_0^{2\pi} d\theta \int_0^R q\, dq \left(-\frac{v^2\Delta}{2(v^2q^2 + \Delta^2)^{3/2}}\right) \tag{43}$$

Solving this integral yields:

$$\gamma_C = -\pi\left[1 - \frac{\Delta}{\sqrt{v^2R^2 + \Delta^2}}\right] \tag{44}$$

#### 6.4.4. The Quantized Result

The final step is to consider the physical situation at the Dirac Nodal Line, which is defined by the absence of a gap, meaning $\Delta = 0$. Taking the limit of our result as $\Delta \to 0$:

$$\gamma_C = \lim_{\Delta\to 0}\left(-\pi\left[1 - \frac{\Delta}{\sqrt{v^2R^2 + \Delta^2}}\right]\right) = -\pi[1 - 0] = -\pi \tag{45}$$

This demonstrates that any loop enclosing the DNL acquires a quantized Berry phase of $-\pi$ (or $+\pi$, depending on the chirality of the Dirac point). This quantized, non-zero value is the topological invariant that protects the nodal line, signifying a non-trivial winding of the quantum state in momentum space.

## 7. Absence of the Magnon Spin Nernst Effect in the DNL Phase

A key prediction of the paper is that the magnon spin Nernst effect, which might be expected in a 2D honeycomb antiferromagnet with DMI, is strictly absent in the layered 3D system in the DNL phase ($H = 0$). This absence is not an approximation but a fundamental consequence of the system's underlying symmetry.

### 7.1. The Magnon Spin Nernst Effect and its Relation to Berry Curvature

The **Magnon spin Nernst effect** is a transverse transport phenomenon. When a temperature gradient is applied across a magnetic material (e.g., along the x-axis, $-\nabla_x T$), it can generate a spin current that flows perpendicularly to both the gradient and the spin polarization (e.g., along the y-axis, $J^s_y$). This is the magnonic analogue of the electronic spin Nernst effect.

The magnitude of this effect is quantified by the spin Nernst conductivity, $\alpha^s_{xy}$. Crucially, in the linear response regime, this conductivity is directly proportional to the integral of the **Berry curvature**, $\Omega_{xy}(\mathbf{k})$, over the Brillouin zone. The relationship is:

$$\alpha^s_{xy} \propto \int_{BZ} d^3k \sum_n g(E_n(\mathbf{k}))\Omega^n_{xy}(\mathbf{k}) \tag{46}$$

where $g(E)$ is a function related to the Bose-Einstein distribution. This relationship provides a clear path to proving the absence of the effect: if we can prove that the Berry curvature $\Omega^n_{xy}(\mathbf{k})$ is identically

zero for all bands $n$ and all momenta $\mathbf{k}$, then the spin Nernst conductivity must also be zero.

### 7.2. Mathematical Derivation of Vanishing Berry Curvature

We will now explicitly show that the Berry curvature is zero by using the mathematical formalism provided in the paper and its supplementary material.

#### 7.2.1. The Formula for Magnon Berry Curvature

As given in Equation (11) of the main paper, the Berry curvature for a bosonic system described by a paraunitary matrix $\mathcal{U}_k$ is:

$$[\Omega_{\alpha\beta}(\mathbf{k})]_{nn} = -2\Im\left[\left(\tau_3\frac{\partial\mathcal{U}_k^\dagger}{\partial k_\alpha}\right)\left(\tau_3\frac{\partial\mathcal{U}_k}{\partial k_\beta}\right)\right]_{nn} \tag{47}$$

Here, $\Im$ denotes the imaginary part, $n$ is the band index, and $\tau_3 = \mathrm{diag}(1,1,-1,-1)$ is the Pauli matrix in the particle-hole space.

#### 7.2.2. The Paraunitary Matrix for the DNL Phase

The supplementary material provides the explicit form of the paraunitary matrix $\mathcal{U}_k$ that diagonalizes the Hamiltonian in the DNL phase (H=0). From Eqs. (21), (25), and (26) of the supplementary material, we have:

$$\mathcal{U}_k = \begin{pmatrix} u_k & v_k \\ v_k & u_k \end{pmatrix} \tag{48}$$

where the $2\times 2$ blocks $u_k$ and $v_k$ are given by:

$$u_k = \frac{1}{\sqrt{2}}\begin{pmatrix} -e^{i\varphi_{k_{||}}}\cosh\phi_k & e^{i\varphi_{k_{||}}}\cosh\theta_k \\ -\cosh\phi_k & -\cosh\theta_k \end{pmatrix} \tag{49}$$

$$v_k = \frac{1}{\sqrt{2}}\begin{pmatrix} e^{i\varphi_{k_{||}}}\sinh\phi_k & -e^{i\varphi_{k_{||}}}\sinh\theta_k \\ \sinh\phi_k & \sinh\theta_k \end{pmatrix} \tag{50}$$

The functions $\phi_k$, $\theta_k$, and $\varphi_{k_{||}}$ are real functions of the momentum $\mathbf{k}$.

#### 7.2.3. Step-by-Step Calculation

Let's compute the terms in the Berry curvature formula. First, the product $\tau_3\mathcal{U}_k^\dagger$ is:

$$\tau_3\mathcal{U}_k^\dagger = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}\begin{pmatrix} u_k^\dagger & v_k^\dagger \\ v_k^\dagger & u_k^\dagger \end{pmatrix} = \begin{pmatrix} u_k^\dagger & v_k^\dagger \\ -v_k^\dagger & -u_k^\dagger \end{pmatrix} \tag{51}$$

Similarly, for the second part of the formula:

$$\tau_3\frac{\partial\mathcal{U}_k}{\partial k_\beta} = \begin{pmatrix} I & 0 \\ 0 & -I \end{pmatrix}\begin{pmatrix} \partial_\beta u_k & \partial_\beta v_k \\ \partial_\beta v_k & \partial_\beta u_k \end{pmatrix} = \begin{pmatrix} \partial_\beta u_k & \partial_\beta v_k \\ -\partial_\beta v_k & -\partial_\beta u_k \end{pmatrix} \tag{52}$$

Now we multiply the two resulting matrices. Let's call the product matrix $M$:

$$M = \left(\tau_3\frac{\partial\mathcal{U}_k^\dagger}{\partial k_\alpha}\right)\left(\tau_3\frac{\partial\mathcal{U}_k}{\partial k_\beta}\right) = \begin{pmatrix} \partial_\alpha u_k^\dagger\partial_\beta u_k - \partial_\alpha v_k^\dagger\partial_\beta v_k & \dots \\ \dots & \dots \end{pmatrix} \tag{53}$$

The crucial step is to analyze the diagonal blocks of this product. The paraunitarity condition $\mathcal{U}_k^\dagger\tau_3\mathcal{U}_k = \tau_3$ implies the relation $u_k^\dagger u_k - v_k^\dagger v_k = I$. Differentiating this identity with respect to a momentum component $k_\beta$ gives:

$$(\partial_\beta u_k^\dagger)u_k + u_k^\dagger(\partial_\beta u_k) - (\partial_\beta v_k^\dagger)v_k - v_k^\dagger(\partial_\beta v_k) = 0 \tag{54}$$

This identity, along with the explicit forms of $u_k$ and $v_k$, can be used to show that the diagonal elements of the matrix product $M$ are **purely real.** As shown explicitly in Eq. (12) of the paper, the imaginary parts of the terms cancel out. Since the diagonal elements are real, their imaginary part is zero:

$$\Im[M_{nn}] = 0 \tag{55}$$

Therefore, the Berry curvature for every band is identically zero:

$$[\Omega_{\alpha\beta}(\mathbf{k})]_{nn} = -2\times\Im[M_{nn}] = 0 \tag{56}$$

### 7.3. Physical Interpretation: The Role of PT Symmetry

The mathematical result that $\Omega(\mathbf{k}) = 0$ is a direct consequence of the underlying **PT symmetry** of the DNL phase. As explained previously, PT symmetry imposes the strict constraint that the Berry curvature must be zero everywhere. This calculation explicitly demonstrates how that symmetry manifests in the mathematical formalism of the paraunitary matrix. Since the Berry curvature is zero, the integral for the spin Nernst conductivity is also zero, and thus the **magnon spin Nernst effect is absent** in this system.

## 8. Analysis of NNN Interlayer Spin Canting Interaction on Dirac Nodal Lines

In this section, we extend the model presented by S. A. Owerre[1] for layered honeycomb antiferromagnets to include a next-nearest-neighbor (NNN) Dzyaloshinskii-Moriya interaction (DMI) between the layers. We will derive the magnon energy spectrum and demonstrate analytically why this interaction, despite modifying the bands, fails to open a gap in the Dirac Nodal Line (DNL) that exists along the K-H high-symmetry path at $k_z = \pi/2$.

### 8.1. Spin Hamiltonian with NNN Interlayer Spin Canting Interaction

We begin with the spin Hamiltonian for a layered honeycomb lattice with a collinear Néel antiferromagnetic ordering. The model includes intralayer nearest-neighbor (NN) Heisenberg exchange ($J$), intralayer next-nearest-neighbor (NNN) DMI ($\mathbf{D}$), interlayer NN Heisenberg coupling ($J_c$), and the newly introduced NNN interlayer spin canting interaction ($\mathbf{D}_c$).

The total Hamiltonian is given by:

$$\mathcal{H} = \mathcal{H}_J + \mathcal{H}_D + \mathcal{H}_{J_c} + \mathcal{H}_{D_{c,NNN}} \tag{57}$$

where the components are:

$$\mathcal{H}_J = J\sum_{\langle i,j\rangle,l}\mathbf{S}_{i,l}\cdot\mathbf{S}_{j,l} \tag{58}$$

$$\mathcal{H}_D = \sum_{\langle\langle i,j\rangle\rangle,l}\mathbf{D}_{ij,l}\cdot(\mathbf{S}_{i,l}\times\mathbf{S}_{j,l}) \tag{59}$$

$$\mathcal{H}_{J_c} = J_c\sum_{\langle l,l'\rangle,i}\mathbf{S}_{i,l}\cdot\mathbf{S}_{i,l'} \tag{60}$$

$$\mathcal{H}_{D_{c,NNN}} = \sum_{\langle i,j\rangle,\langle l,l'\rangle}\mathbf{D}_c\cdot(\mathbf{S}_{i,l}\times\mathbf{S}_{j,l'}) \tag{61}$$

Here, $\mathbf{S}_{i,l}$ is the spin operator at site $i$ in layer $l$. For simplicity, we assume the DMI vectors $\mathbf{D}$ and $\mathbf{D}_c$ are oriented along the stacking direction, i.e., $\mathbf{D} = D\hat{z}$ and $\mathbf{D}_c = D_c\hat{z}$. The term $\mathcal{H}_{D_{c,NNN}}$ describes an interaction between a spin on site $i$ (sublattice A) in layer $l$ and a neighboring spin on site $j$ (sublattice B) in an adjacent layer $l'$.

### 8.2. Action of the NNN Interlayer Spin Canting Interaction in terms of Symmetry Breaking

The NNN interlayer spin canting interaction, represented by $\mathcal{H}_{D_{c,NNN}}$, introduces a new type of Dzyaloshinskii-Moria interaction that couples spins in adjacent layers. This term is distinct from the intralayer DMI, $\mathcal{H}_D$, which acts only within a single layer. The key features of this interaction are:

- **Symmetry Considerations:** The NNN interlayer DMI-like breaks certain symmetries of the original Hamiltonian. While the intralayer DMI preserves the combined PT symmetry, the interlayer DMI can break inversion symmetry between layers. However, it does not break the overall PT symmetry of the system, which is crucial for the protection of the Dirac Nodal Lines.

- **Effect on Spin Configuration:** The NNN interlayer DMI-like induces a canting of spins between adjacent layers. This canting modifies the effective magnetic interactions and can lead to non-collinear spin arrangements. However, in the collinear Néel state considered here, the primary effect is to introduce additional terms in the magnon Hamiltonian that couple the layers.

### 8.3. Bosonization and Momentum Space Representation

At low temperatures, we employ the Holstein-Primakoff transformation to describe the magnon excitations. For the two-sublattice Néel state, we define bosonic operators $a$ for sublattice A and $b$ for sublattice B:

$$\text{Sublattice A: } S^z_{i,l} = S - a^\dagger_{i,l}a_{i,l}, \quad S^+_{i,l} \approx \sqrt{2S}a_{i,l} \tag{62}$$

$$\text{Sublattice B: } S^z_{j,l} = -S + b^\dagger_{j,l}b_{j,l}, \quad S^+_{j,l} \approx \sqrt{2S}b^\dagger_{j,l} \tag{63}$$

After applying a Fourier transform, the quadratic part of the Hamiltonian can be written in matrix form:

$$\mathcal{H} = \frac{1}{2}\sum_{\mathbf{k}} \Psi^\dagger(\mathbf{k})\mathcal{H}(\mathbf{k})\Psi(\mathbf{k}) \tag{64}$$

where the basis vector is $\Psi^\dagger(\mathbf{k}) = (a^\dagger_{\mathbf{k}}, b^\dagger_{\mathbf{k}}, a_{-\mathbf{k}}, b_{-\mathbf{k}})$. The Hamiltonian matrix takes the Bogoliubov-de Gennes (BdG) form:

$$\mathcal{H}(\mathbf{k}) = \begin{pmatrix} \mathcal{A}(\mathbf{k}) & \mathcal{B}(\mathbf{k}) \\ \mathcal{B}^*(-\mathbf{k}) & \mathcal{A}^*(-\mathbf{k}) \end{pmatrix} \tag{65}$$

The NNN interlayer spin canting term introduces new off-diagonal elements into the $\mathcal{A}(\mathbf{k})$ block. The resulting blocks are:

$$\mathcal{A}(\mathbf{k}) = \begin{pmatrix} f_0 + f^D(\mathbf{k}_{||}) & if^{D_c}(\mathbf{k}) \\ -if^{D_c*}(\mathbf{k}) & f_0 - f^D(\mathbf{k}_{||}) \end{pmatrix} \tag{66}$$

$$\mathcal{B}(\mathbf{k}) = \begin{pmatrix} f(\mathbf{k}_z) & f(\mathbf{k}_{||}) \\ f^*(\mathbf{k}_{||}) & f(\mathbf{k}_z) \end{pmatrix} \tag{67}$$

The functions are defined as:

- $f_0 = 3JS + 2J_cS$
- $f(\mathbf{k}_z) = 2J_cS\cos(k_z)$
- $f(\mathbf{k}_{||}) = JS(1 + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_1} + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_2})$
- $f^D(\mathbf{k}_{||}) = 2DS(\sin(\mathbf{k}_{||}\cdot\mathbf{a}_1) - \sin(\mathbf{k}_{||}\cdot\mathbf{a}_2) + \sin(\mathbf{k}_{||}\cdot(\mathbf{a}_2 - \mathbf{a}_1)))$
- $f^{D_c}(\mathbf{k}) = 2D_cS\cos(k_z)f(\mathbf{k}_{||})/(JS)$

The crucial new term, $f^{D_c}(\mathbf{k})$, arises from the NNN interlayer spin canting interaction. Its dependence on both $\cos(k_z)$ and $f(\mathbf{k}_{||})$ is the key to understanding its effect on the DNLs.

### 8.4. Detailed Derivation of the Hamiltonian Matrix Elements

This section provides a comprehensive, step-by-step derivation of the functions that constitute the momentum-space Hamiltonian matrix, $\mathcal{H}(\mathbf{k})$. Each function is derived by applying the Holstein-Primakoff (HP) transformation to the relevant term in the spin Hamiltonian, followed by a Fourier transform.

#### 8.4.1. Derivation of $f_0$ (On-site Energy Term)

The on-site energy term, $f_0$, originates from the diagonal parts of the Heisenberg exchange interactions after bosonization.

**1. Intralayer Contribution:** We start with the intralayer Heisenberg term, $\mathcal{H}_J = J\sum_{\langle i,j\rangle}\mathbf{S}_{i,l}\cdot\mathbf{S}_{j,l}$. The $S^zS^z$ component for a site $i \in A$ and neighbor $j \in B$ is:

$$\begin{aligned} JS^z_{i,l}S^z_{j,l} &= J(S - a^\dagger_{i,l}a_{i,l})(-S + b^\dagger_{j,l}b_{j,l}) \\ &= -JS^2 + JS(a^\dagger_{i,l}a_{i,l} + b^\dagger_{j,l}b_{j,l}) + \mathcal{O}(a^4) \end{aligned}$$

Each site has three nearest neighbors, so summing over them contributes a term $3JSa^\dagger_{i,l}a_{i,l}$ to the on-site energy of site $i$.

**2. Interlayer Contribution:** From the interlayer term $\mathcal{H}_{J_c} = J_c\sum_{i,\langle l,l'\rangle}\mathbf{S}_{i,l}\cdot\mathbf{S}_{i,l'}$, where spins are antiparallel:

$$\begin{aligned} J_cS^z_{i,l}S^z_{i,l'} &= J_c(S - a^\dagger_{i,l}a_{i,l})(-S + a^\dagger_{i,l'}a_{i,l'}) \\ &= -J_cS^2 + J_cS(a^\dagger_{i,l}a_{i,l} + a^\dagger_{i,l'}a_{i,l'}) + \ldots \end{aligned}$$

Each site has two interlayer neighbors (above and below), contributing $2J_cSa^\dagger_{i,l}a_{i,l}$ to the on-site energy.

**3. Total On-site Term:** Combining both contributions, the total coefficient for the number operator $a^\dagger_{i,l}a_{i,l}$ is $(3JS + 2J_cS)$. This momentum-independent term forms the main diagonal part of the $\mathcal{A}(\mathbf{k})$ matrix.

$$\mathbf{f_0 = 3JS + 2J_cS} \tag{68}$$

#### 8.4.2. Derivation of $f(\mathbf{k}_{||})$ (Intralayer Heisenberg Pairing Term)

This function arises from the in-plane components of $\mathcal{H}_J$ and describes magnon pairing within a layer.

**1. Bosonization:** The in-plane interaction is written using ladder operators for $i \in A, j \in B$:

$$\begin{aligned} J(S^x_{i,l}S^x_{j,l} + S^y_{i,l}S^y_{j,l}) &= \frac{J}{2}(S^+_{i,l}S^-_{j,l} + S^-_{i,l}S^+_{j,l}) \\ &\approx \frac{J}{2}\left[(\sqrt{2S}a_{i,l})(\sqrt{2S}b_{j,l}) + (\sqrt{2S}a^\dagger_{i,l})(\sqrt{2S}b^\dagger_{j,l})\right] \\ &= JS(a_{i,l}b_{j,l} + a^\dagger_{i,l}b^\dagger_{j,l}) \end{aligned}$$

This pairing term belongs in the $\mathcal{B}(\mathbf{k})$ block.

**2. Fourier Transform:** We transform the term $\sum_{\langle i,j\rangle}JSa_{i,l}b_{j,l}$, summing over nearest neighbors $\boldsymbol{\delta}_\nu$:

$$\begin{aligned} \sum_{i,\nu}JSa_{i,l}b_{i+\delta_\nu,l} &= \sum_{i,\nu}JS\left(\frac{1}{\sqrt{N}}\sum_{\mathbf{k}}e^{i\mathbf{k}\cdot\mathbf{r}_i}a_{\mathbf{k},l}\right)\left(\frac{1}{\sqrt{N}}\sum_{\mathbf{k}'}e^{i\mathbf{k}'\cdot(\mathbf{r}_i+\delta_\nu)}b_{\mathbf{k}',l}\right) \\ &= \sum_{\mathbf{k},\mathbf{k}'}JS\,\delta_{\mathbf{k},-\mathbf{k}'}\,a_{\mathbf{k},l}b_{\mathbf{k}',l}\left(\sum_\nu e^{i\mathbf{k}'\cdot\delta_\nu}\right) \\ &= \sum_{\mathbf{k}}JS\left(\sum_\nu e^{-i\mathbf{k}\cdot\delta_\nu}\right)a_{\mathbf{k},l}b_{-\mathbf{k},l} \end{aligned}$$

The geometric factor for the honeycomb lattice is $(1 + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_1} + e^{-i\mathbf{k}_{||}\cdot\mathbf{a}_2})$.

$$\mathbf{f(k_{||}) = JS(1 + e^{-ik_{||}\cdot a_1} + e^{-ik_{||}\cdot a_2})} \tag{69}$$

#### 8.4.3. Derivation of $f(k_z)$ (Interlayer Heisenberg Pairing Term)

This function describes magnon pairing between vertically aligned sites and comes from $\mathcal{H}_{J_c}$.

**1. Bosonization:** For site $i$ on layer $l$ (A, up) and $l'$ (A, down):

$$\frac{J_c}{2}(S^+_{i,l}S^-_{i,l'} + S^-_{i,l}S^+_{i,l'}) \approx J_cS(a_{i,l}a_{i,l'} + a^\dagger_{i,l}a^\dagger_{i,l'})$$

**2. Fourier Transform:** Summing over adjacent layers ($l' = l \pm 1$) and transforming with respect to the layer index gives:

$$\begin{aligned} \sum_l J_cS(a^\dagger_{\mathbf{k},l}a^\dagger_{-\mathbf{k},l+1} + a^\dagger_{\mathbf{k},l}a^\dagger_{-\mathbf{k},l-1}) &\to J_cS(e^{ik_z} + e^{-ik_z})a^\dagger_{\mathbf{k},k_z}a^\dagger_{-\mathbf{k},-k_z} \\ &= 2J_cS\cos(k_z)a^\dagger_{\mathbf{k},k_z}a^\dagger_{-\mathbf{k},-k_z} \end{aligned}$$

This same-sublattice pairing term is on the diagonal of the $\mathcal{B}(\mathbf{k})$ block.

$$\mathbf{f(k_z) = 2J_cS\cos(k_z)} \tag{70}$$

#### 8.4.4. Derivation of $f^D(\mathbf{k}_{||})$ (Intralayer DMI Term)

This function arises from the intralayer NNN DMI, $\mathcal{H}_D = \sum_{\langle\langle i,j\rangle\rangle}\nu_{ij}D(\mathbf{S}_i \times \mathbf{S}_j)_z$.

**1. Bosonization:** NNN sites are on the same sublattice (e.g., A).

The DMI-like term becomes:

$$\nu_{ij}D(S_i^xS_j^y - S_i^yS_j^x) = \frac{\nu_{ij}D}{2i}(S_i^+S_j^- - S_i^-S_j^+)$$
$$\approx -i\nu_{ij}DS(a_ia_j^\dagger - a_i^\dagger a_j)$$

This is a hopping term between NNN sites and belongs in the $\mathcal{A}(\mathbf{k})$ block.

**2. Fourier Transform:** Transforming the term $\sum_{\langle\langle i,j\rangle\rangle} i\nu_{ij}DSa_i^\dagger a_j$ yields a momentum-dependent coefficient from the NNN geometric factor, which gives:

$$\mathbf{f^D}(\mathbf{k}_{||}) = \mathbf{2DS}(\sin(\mathbf{k}_{||}\cdot\mathbf{a_1}) - \sin(\mathbf{k}_{||}\cdot\mathbf{a_2}) + \sin(\mathbf{k}_{||}\cdot(\mathbf{a_2}-\mathbf{a_1}))) \tag{71}$$

This term adds to the diagonal of $\mathcal{A}(\mathbf{k})$ with opposite signs for the two sublattices.

#### *8.4.5. Derivation of $f^{D_c}(\mathbf{k})$ (NNN Interlayer spin canting interaction Term)*

This function describes magnon hopping between different sublattices on adjacent layers.

**1. Bosonization:** The term is $\mathcal{H}_{D_c} = \sum_{\langle i,j\rangle,\langle l,l'\rangle} D_c(\mathbf{S}_{i,l}\times\mathbf{S}_{j,l'})_z$. Here, site $i\in A$ on layer $l$ is spin-up, and its neighbor $j\in B$ on layer $l'$ is also spin-up.

$$D_c(S_{i,l}^xS_{j,l'}^y - S_{i,l}^yS_{j,l'}^x) = \frac{D_c}{2i}(S_{i,l}^+S_{j,l'}^- - S_{i,l}^-S_{j,l'}^+)$$
$$\approx -iD_cS(a_{i,l}b_{j,l'}^\dagger - a_{i,l}^\dagger b_{j,l'})$$

This is a hopping term that modifies the off-diagonal elements of the $\mathcal{A}(\mathbf{k})$ block.

**2. Fourier Transform:** We transform the term $\sum iD_cSa_{i,l}^\dagger b_{j,l'}$. The sum is over NN bonds $\boldsymbol{\delta}$ and adjacent layers ($l' = l\pm 1$).

$$\sum_{i,\delta,l,\pm} iD_cSa_{i,l}^\dagger b_{i+\delta,l\pm1} \to \sum_{\mathbf{k}} iD_cS\left(\sum_\delta e^{-i\mathbf{k}_{||}\cdot\delta}\right)(e^{ik_z}+e^{-ik_z})a_{\mathbf{k}}^\dagger b_{\mathbf{k}}$$
$$= i\cdot 2D_cS\cos(k_z)\frac{f^*(\mathbf{k}_{||})}{JS}a_{\mathbf{k}}^\dagger b_{\mathbf{k}}$$

This coefficient becomes the $\mathcal{A}_{12}(\mathbf{k})$ element of the Hamiltonian matrix. We define its magnitude as:

$$\mathbf{f^{D_c}}(\mathbf{k}) = \mathbf{2D_cS}\cos(\mathbf{k_z})\frac{|\mathbf{f}(\mathbf{k}_{||})|}{\mathbf{JS}} \tag{72}$$

The matrix element itself is $if^{D_c}(\mathbf{k})\cdot e^{-i\phi_k}$, where $\phi_k$ is the phase of $f(\mathbf{k}_{||})$.

### 8.5. Loss of the DNL Phase due to Interlayer Spin-Canting Interaction

The plot below illustrates the consequences of this analytical result. The green dotted lines, representing the model with the NNN interlayer spin-canting interaction ($D_c$), show a clear gap opening along paths like K-H. This confirms that the interaction breaks the protecting PT symmetry and destroys the Dirac Nodal Line phase present in the original models.

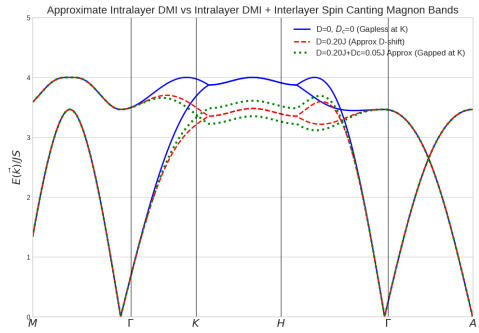


**Figure 3.** Magnon dispersion bands comparing the original model (blue and red lines) with the extended model including the NNN interlayer DMI (green dotted lines). The new interaction opens a gap between the magnon bands, indicating the loss of the DNL phase.

### 8.6. Stability of the DNL at $k_z = \pi/2$

The existence of DNLs in the original model (without NNN interlayer Spin canting interaction) is guaranteed when the two magnon branches become degenerate. This occurs when the argument of the square root in the energy dispersion vanishes, which corresponds to the condition[1]:

$$f_0^2 - [f(k_z) \pm |f(k_{||})|]^2 = 0 \tag{73}$$

This equation is satisfied along lines in the Brillouin zone where either $f(k_z) = 0$ or $|f(k_{||})| = 0$. The high-symmetry path from K to H is defined by $\mathbf{k}_{||} = \mathbf{K} = (4\pi/3\sqrt{3}, 0)$ while $k_z$ varies. A DNL exists on this path where the bands cross.

Now, let us analyze the effect of our new interaction, $\mathcal{H}_{D_{c,NNN}}$, specifically in the plane where $k_z = \pi/2$.

**Step 1: Evaluate the new Hamiltonian term at $k_z = \pi/2$.** The entire contribution from the NNN interlayer spin canting interaction is contained in the function $f^{D_c}(\mathbf{k})$. Let us evaluate this function at $k_z = \pi/2$:

$$f^{D_c}(\mathbf{k}_{||}, k_z = \pi/2) = 2D_cS\cos(\pi/2)\frac{f(\mathbf{k}_{||})}{JS} \tag{74}$$

Since $\cos(\pi/2) = 0$, the function vanishes completely:

$$f^{D_c}(\mathbf{k}_{||}, k_z = \pi/2) = 0 \tag{75}$$

**Step 2: Examine the Hamiltonian Matrix at $k_z = \pi/2$.** Substituting this result back into the Hamiltonian matrix $\mathcal{H}(\mathbf{k})$, the $\mathcal{A}(\mathbf{k})$ block becomes diagonal:

$$\mathcal{A}(\mathbf{k}_{||}, k_z = \pi/2) = \begin{pmatrix} f_0 + f^D(\mathbf{k}_{||}) & 0 \\ 0 & f_0 - f^D(\mathbf{k}_{||}) \end{pmatrix} \tag{76}$$

Simultaneously, the interlayer Heisenberg term in the $\mathcal{B}(\mathbf{k})$ block also vanishes:

$$f(k_z = \pi/2) = 2J_cS\cos(\pi/2) = 0 \tag{77}$$

This simplifies the $\mathcal{B}(\mathbf{k})$ block to:

$$\mathcal{B}(\mathbf{k}_{||}, k_z = \pi/2) = \begin{pmatrix} 0 & f(\mathbf{k}_{||}) \\ f^*(\mathbf{k}_{||}) & 0 \end{pmatrix} \tag{78}$$

**Step 3: Conclusion of the Mathematical Analysis.** At $k_z = \pi/2$, the Hamiltonian matrix becomes identical to the Hamiltonian of a single, isolated 2D honeycomb layer. The NNN interlayer spin canting, just like the NN interlayer Heisenberg coupling, has absolutely no effect on the magnon dynamics within this specific momentum plane. Since the Hamiltonian reduces to the 2D case, its spectrum must also be that of the 2D case, which is known to host Dirac points at the K and K' points. The line connecting these points as $k_z$

varies forms the DNL. Because the interaction vanishes at $k_z = \pi/2$, it cannot lift the degeneracy and open a gap in that plane. The DNL is therefore robust against this specific interaction.

### 8.7. Symmetry Protection of the Nodal Line

The stability of the DNL can also be understood from the perspective of the combined PT (inversion and time-reversal) symmetry. The collinear Néel state is invariant under PT. Any interaction added to the Hamiltonian must break this protecting symmetry to gap out the DNL.

The NNN interlayer spin canting term, $\mathcal{H}_{D_{c,NNN}}$, is odd under the inversion operation P (which sends $k_z \to -k_z$) because of the $\cos(k_z)$ factor, but it is even under time-reversal T. Therefore, the term is odd under the combined PT operation and breaks the protecting symmetry.

However, this symmetry-breaking argument is subtle. While the interaction term itself breaks PT symmetry globally, its effect *vanishes* at the specific $k_z = \pi/2$ plane. At this plane, the Hamiltonian effectively recovers the higher symmetry of the 2D system. The symmetry protection of the nodal line is restored within this 2D subspace of the Brillouin zone because the symmetry-breaking field (the NNN interlayer spin canting) has zero magnitude there. Therefore, while the DNL may be gapped elsewhere, it remains protected and gapless along any path that lies entirely within the $k_z = \pi/2$ plane, such as the line connecting the H points in the Brillouin zone.

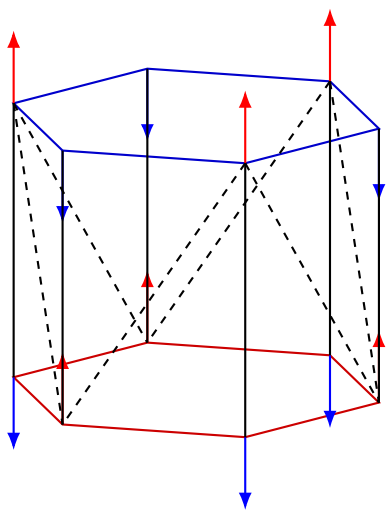

**Figure 4.** A 3D visualization of the stacked honeycomb lattice with a collinear Néel antiferromagnetic (AFM) spin configuration. The solid vertical lines represent the nearest-neighbor interlayer coupling ($J_c$), while the dashed lines indicate the next-nearest-neighbor interlayer spin canting interaction paths.

## 9. Conclusion and Further Discussion

In summary, our study reveals that the stability of Dirac Nodal Lines in layered honeycomb antiferromagnets is intricately linked to the presence of PT symmetry and the specific momentum dependence of interlayer interactions. While the introduction of a NNN interlayer spin canting term generically gaps out the nodal lines by breaking PT symmetry, the DNL persists in the $k_z = \pi/2$ plane due to the vanishing of the interaction at this momentum slice. This demonstrates that topological protection can be locally restored in certain regions of the Brillouin zone, even when global symmetry is broken. These results provide deeper insight into the interplay between symmetry, topology, and magnon band structure, and suggest possible routes for engineering robust topological magnonic phases through careful control of interlayer couplings.

## References

[1] S. Owerre, "Topological magnon nodal-lines and absence of magnon spin nernst effect in layered collinear antiferromagnets", *EPL (Europhysics Letters)*, vol. 125, no. 3, p. 36 002, 2019.

## List of Figures

1 The NNN DMI path between two sites $i$ and $j$ on the same sublattice (blue). The sign $\nu_{ij}$ is determined by the intermediate site $m$ (magenta). For a counter-clockwise path $i \to m \to j$, $\nu_{ij} = +1$, while for a clockwise path, it would be $-1$. . . . . 3
2 Magnon dispersion bands along high-symmetry lines in the Brillouin zone for $J_c/J = 0.5$. The solid blue lines correspond to $D = 0$, showing Dirac Nodal Lines (DNLs) where the bands touch. The dashed red lines ($D = 0.2$) show that the DNLs persist even with DMI. 5
3 Magnon dispersion bands comparing the original model (blue and red lines) with the extended model including the NNN interlayer DMI (green dotted lines). The new interaction opens a gap between the magnon bands, indicating the loss of the DNL phase. . . . 9
4 A 3D visualization of the stacked honeycomb lattice with a collinear Néel antiferromagnetic (AFM) spin configuration. The solid vertical lines represent the nearest-neighbor interlayer coupling ($J_c$), while the dashed lines indicate the next-nearest-neighbor interlayer spin canting interaction paths. . . . . 10